

Role of antioxidants in cancer therapy.

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Oxidative stress is a key component in linking environmental toxicity to the multistage carcinogenic process. Reactive oxygen species (ROS) are generated in response to both endogenous and exogenous stimuli. To counterbalance ROS-mediated injury, an endogenous antioxidants defense system exists; however, when oxidation exceeds the control mechanisms, oxidative stress arises. Chronic and cumulative oxidative stress induces deleterious modifications to a variety of macromolecular components, such as DNA, lipids, and proteins. A primary mechanism of many chemotherapy drugs against cancer cells is the formation of ROS, or free radicals. Radiotherapy is based on the fact that ionizing radiation destroys tumor cells. Radiotherapy induces direct lesions in the DNA or biological molecules, which eventually affect DNA. Free radicals produced by oncology therapy are often a source of serious side effects as well. The objective of this review is to provide information about the effects of antioxidants during oncology treatments and to discuss the possible events and efficacy. Much debate has arisen about whether antioxidant supplementation alters the efficacy of cancer chemotherapy. There is still limited evidence in both quality and sample size, suggesting that certain antioxidant supplements may reduce adverse reactions and toxicities. Significant reductions in toxicity may alleviate dose-limiting toxicities so that more patients are able to complete prescribed chemotherapy regimens and thus, in turn, improve the potential for success in terms of tumor response and survival. Copyright © 2013 Elsevier Inc. All rights reserved.